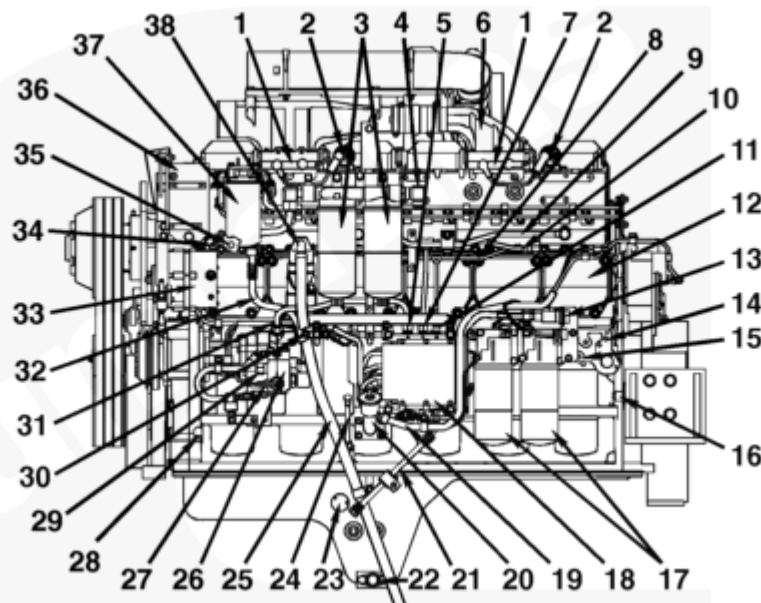


Trainings ▾

Engine Views

The illustrations show the locations of the major external engine components, filters, and other service and maintenance points. Some external components will be at different locations for different engine models.

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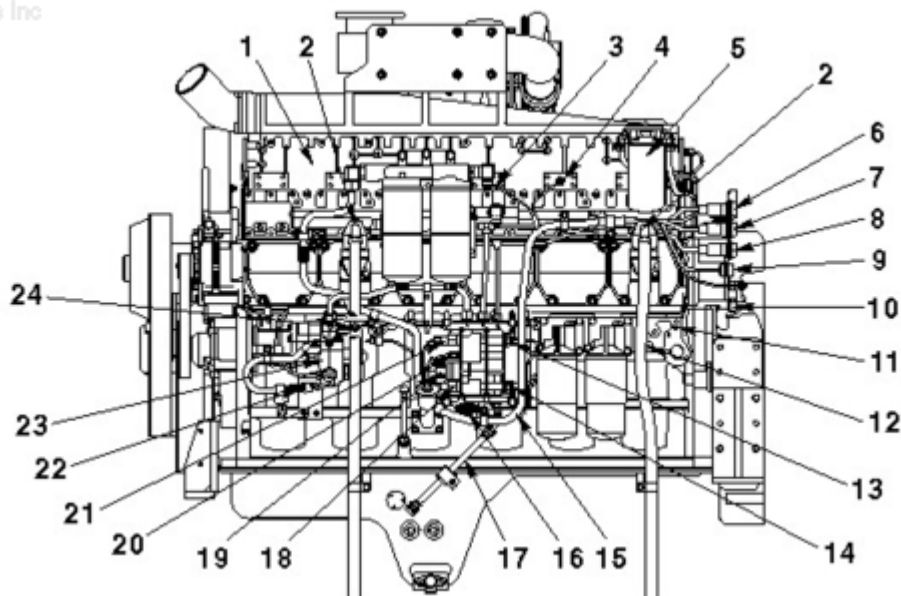
00400154

Left Side - Industrial

1. Grid heaters (Industrial **only**)
2. Grid heater relays (Industrial **only**)
3. Fuel filters
4. Fuel inlet (to filters)
5. Fuel inlet (to ECVA)
6. Air intake connection (Industrial **only**)
7. Fuel timing supply
8. Fuel pipe (to fuel rail)
9. Fuel pipe (return)
10. Fuel pipe (for injection timing)
11. Fuel rail supply
12. Cam follower cover
13. OEM electrical connections (Industrial **only**)
14. Oil pressure checkpoint (after filters)
15. Oil pressure checkpoint (before filters)

16. Engine barring device
17. Oil filters
18. ECVA/ECM
19. Wiring harness
20. Oil fill port
21. Fuel rail dampening hose
22. Engine oil drain valve
23. Oil level sensor (optional)
24. Dipstick
25. Blowby tube
26. Fuel pump
27. Fuel pump pressure sensor
28. Oil pressure check point (to front idler gears)
29. Fuel pump actuator
30. Oil pressure sensor
31. Fuel pipe (pump to ECVA)
32. Fuel pipe (filters to pump)
33. Alternator
34. Intake manifold pressure sensor (Industrial **only**)
35. Intake manifold temperature sensor (Industrial **only**)
36. Refrigerant compressor mounting location
37. Coolant filter (Industrial **only**)
38. Crankcase breather assembly.

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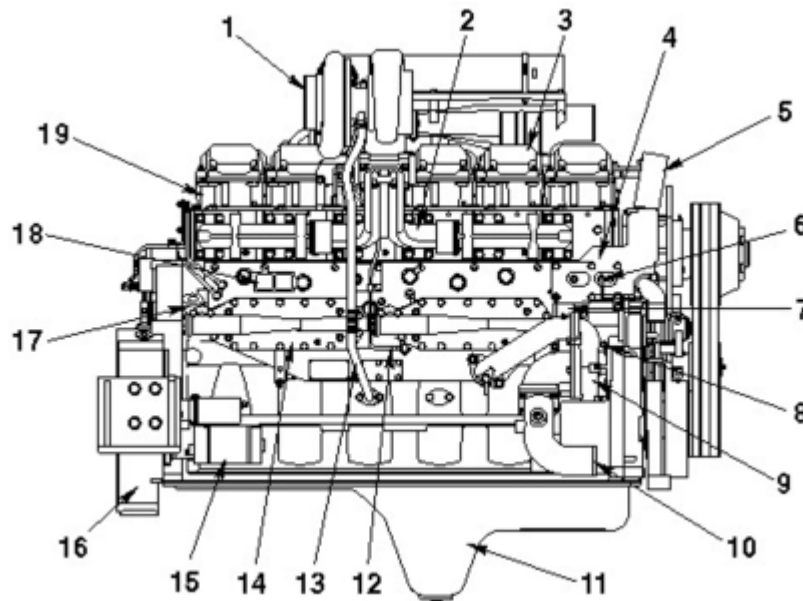
00400162

Left Side - Power Generation

1. Intake manifold (Power Generation **only**)
2. Crankcase breathers
3. Intake manifold pressure sensor (Power Generation **only**)
4. Intake manifold temperature sensor (Power Generation **only**)
5. Coolant filter (Power Generation **only**)

6. Inline B connector (Power Generation **only**)
7. Inline A connector (Power Generation **only**)
8. Inline C connector (Power Generation **only**)
9. Datalink connector (Power Generation **only**)
10. Engine speed sensor (Power Generation **only**)
11. Oil pressure checkpoint (after filters)
12. Oil pressure checkpoint (before filters)
13. Fuel timing pressure sensor
14. Fuel rail pressure sensor
15. Wiring harness
16. Barometric pressure sensor
17. Fuel rail dampening hose
18. Fuel rail actuator
19. Fuel shutoff valve
20. Fuel temperature sensor
21. Fuel timing actuator
22. Fuel pump pressure sensor
23. Fuel pump actuator
24. Oil pressure sensor.

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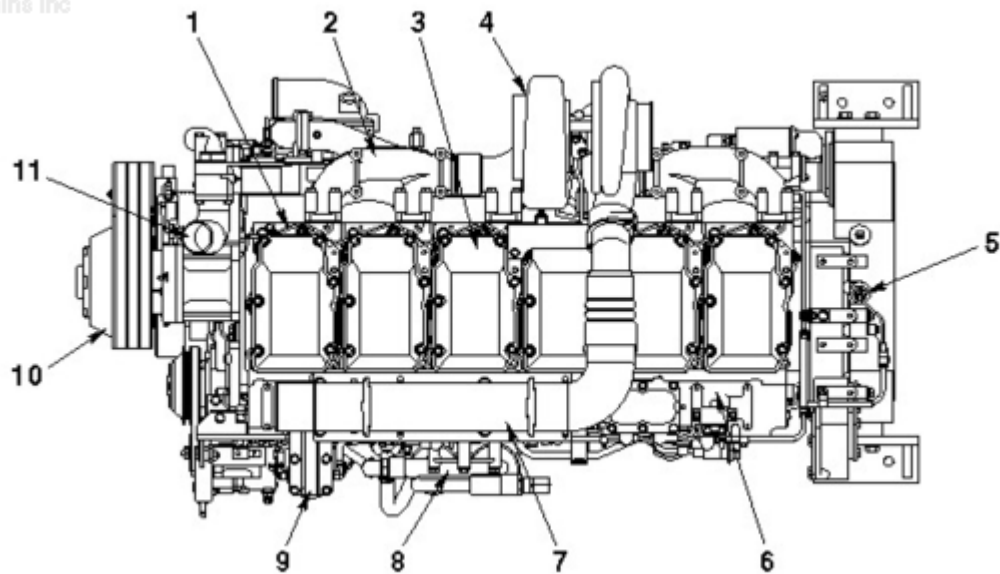


00400165

Right Side - Industrial and Power Generation

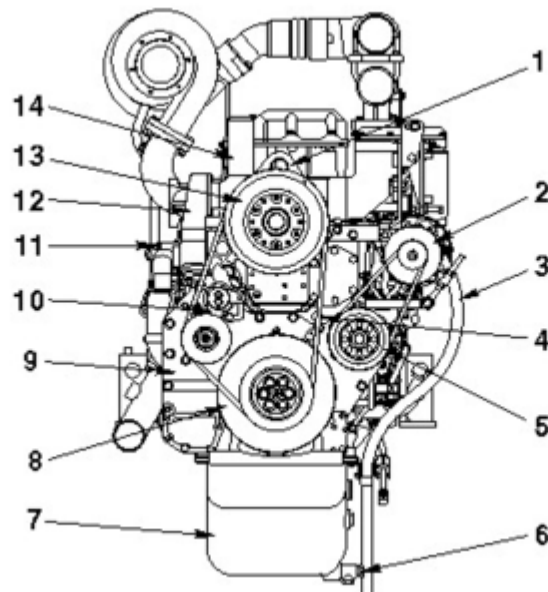
1. Turbocharger
2. Exhaust manifold (3-piece)
3. Rocker lever cover
4. Thermostat housing
5. Engine coolant outlet
6. Coolant temperature sensor
7. Coolant bypass tube
8. Oil pressure checkpoint (piston cooling nozzle gallery)
9. Water pump

10. Engine coolant inlet
11. Oil pan
12. Turbocharger oil supply line
13. Turbocharger oil drain line
14. Oil cooler housing
15. Starter
16. Flywheel housing
17. Oil pressure checkpoint (after oil coolers)
18. Engine dataplate
19. Rocker lever housing.



Top View - Industrial and Power Generation

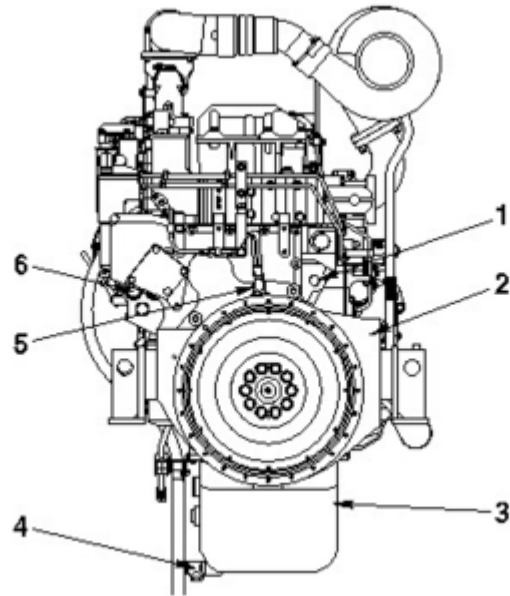
1. Coolant vent tube
2. Exhaust manifold
3. Rocker lever cover
4. Turbocharger
5. Engine speed sensor (Industrial **only**)
6. Intake manifold
7. Charge air piping (turbocharger to CAC)
8. Fuel filter head
9. Coolant filter head (Industrial **only**)
10. Fan hub assembly
11. Engine coolant outlet.



00400167

Front View - Industrial and Power Generation

1. Front engine lifting bracket
2. Alternator
3. Blowby tube
4. Cooling fan drive belt
5. Drive pulley (for alternator and refrigerant compressor)
6. Oil drain valve
7. Oil pan
8. Vibration damper
9. Front gear cover
10. Cooling fan belt tensioner (Industrial **only**)
11. Coolant temperature sensor
12. Thermostat housing
13. Fan hub assembly
14. Engine coolant outlet.



00400165

Rear View - Industrial and Power Generation

1. Oil pressure checkpoint (after oil coolers)
2. Flywheel housing
3. Oil pan
4. Engine oil drain valve
5. Engine speed sensor (Industrial **only**)
6. OEM electrical connections (Industrial **only**).